# EV CHARGING NETWORK USAGE ANALYSIS

Microsoft Excel Analytics & Dashboard Case Study

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**Project focus:** Data preparation • Advanced Excel formulas • Statistical analysis • PivotTables • Dashboarding

6,500 charging sessions | ■2.26M revenue | 8 cities

# 1. Project Overview

This project analyses EV charging activity across multiple cities to evaluate revenue performance, energy consumption, station utilisation, vehicle usage and customer experience. The workbook uses Microsoft Excel only and combines formula-driven analysis with PivotTables, visualisations, KPI reporting and dashboard interactivity.

### Business Questions

• How much revenue and energy consumption are generated across charging sessions?

• Which station types and vehicle types are used most frequently?

• How variable is charging-session revenue?

• What relationships exist between charging duration, energy consumption and revenue?

• How can the results be presented in an interactive management dashboard?

| TOTAL SESSIONS | TOTAL REVENUE | AVG ENERGY | AVG RATING |
|---|---|---|---|
| 6,500 | ■2.26M | 30.91 kWh | 2.99 / 5 |

# 2. Analytical Approach

**Data preparation:** checked missing values, numeric validity, date/time consistency and the revenue relationship.

**Calculated columns:** energy level, customer satisfaction level, charging-time category, station code, separate time and date fields.

**Lookup analysis:** city-level average energy consumption and average charging duration using AVERAGEIF and VLOOKUP.

**Statistics:** average, standard deviation, variance, coefficient of variation and correlation.

**Dashboard:** KPI cards, city revenue, vehicle energy usage, station-type distribution, energy trends and a charging-duration vs energy scatter plot.

**Excel techniques demonstrated:** IF, DATE, TIME, AVERAGEIF, VLOOKUP, INDEX-MATCH, MODE, STDEV.S, VAR.S, CORREL, PivotTables, conditional formatting and slicers.

# 3. Key Findings

## Revenue Performance

| City | Sessions | Revenue |
|---|---:|---:|
| Kolkata | 871 | ■308,122.98 |
| Hyderabad | 826 | ■294,720.95 |
| Delhi | 827 | ■280,954.51 |
| Ahmedabad | 816 | ■279,913.62 |
| Pune | 792 | ■278,706.40 |
| Chennai | 803 | ■275,917.56 |
| Bangalore | 788 | ■272,049.21 |
| Mumbai | 777 | ■270,106.44 |

Kolkata was the highest-revenue city at ■308,122.98, while Mumbai was lowest among the eight cities at ■270,106.44.

## Station Utilisation

| Station Type | Sessions | Share |
|---|---:|---:|
| Super Fast Charger | 2,235 | 34.38% |
| Standard Charger | 2,141 | 32.94% |
| Fast Charger | 2,124 | 32.68% |

Super Fast Chargers recorded the highest number of sessions (2,235; 34.38%).

# 4. Statistical Analysis

## Revenue Variation

Revenue CV = **60.45%**. This indicates substantial variation in revenue generated by individual charging sessions.

## Variance Comparison

| Variable | Variance |
| --- | ---: |
| Energy Consumed (kWh) | 281.61 |
| Price per kWh | 6.76 |

Energy consumption shows much greater absolute variability than price per kWh.

## Correlation Analysis

| Variables | Correlation | Interpretation |
| --- | ---: | --- |
| Charging Duration ↔ Energy Consumed | -0.010 | Essentially no linear relationship |
| Energy Consumed ↔ Revenue | 0.900 | Very strong positive relationship |

The strong energy–revenue relationship is consistent with the workbook's revenue validation: Revenue = Energy Consumed × Price per kWh.

# 5. Vehicle & Customer Insights

## Vehicle Type Performance

| Vehicle Type | Sessions | Avg Energy | Revenue |
|---|---|---|---|
| 3W EV | 1,628 | 31.71 kWh | ■583,503.21 |
| 2W EV | 1,650 | 30.31 kWh | ■562,522.07 |
| 4W EV | 1,573 | 31.69 kWh | ■560,716.01 |
| Electric Bus | 1,649 | 30.00 kWh | ■553,750.38 |

**Key finding:** 3W EVs generated the highest vehicle-type revenue (■583,503.21) and had the highest average energy consumption (31.71 kWh/session).

## Customer Experience

Average customer rating was **2.99/5**. Fast Charger ratings were highest at 3.01, while Super Fast Charger ratings were lowest at 2.97. The differences are small, so the overall rating is a more important signal than station-type differences.

## Operational Takeaways

• Monitor Super Fast Charger capacity because it has the highest session volume.

• Investigate customer-experience drivers because the overall rating is around 3/5.

• Use energy consumption as a key revenue driver in operational reporting.

• Review unusually high/low revenue sessions because revenue variability is high.

• Investigate the near-zero duration–energy correlation before making operational assumptions.

# 6. Portfolio Value

This project demonstrates the ability to take a raw operational dataset, structure it for analysis, apply Excel formulas and statistical methods, build PivotTable summaries, and communicate findings through a dashboard. It is particularly relevant for entry-level Data Analyst roles requiring strong Excel, problem-solving and business-analysis skills.

**Recommended resume headline:** EV Charging Network Usage Analysis | Microsoft Excel

**Recommended resume bullet:** Analysed 6,500 EV charging sessions across 8 cities using advanced Excel formulas, PivotTables, statistical analysis and an interactive dashboard; identified ■2.26M revenue, 60.45% revenue variability, a 0.90 energy–revenue correlation and Super Fast Chargers as the most-used station type.

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Portfolio QA note: before publishing, correct the Total Charging Sessions formula so it excludes the header, rename “Distribution of State Types” to “Station Type Distribution”, and remove the blank/1900 trend category.